SCENE-SKDE: A MULTI-AGENT KEYPOINT-DENSITY EXTENSION OF SEEKER FOR PEDESTRIAN–VEHICLE HAZARD DETECTION

CMP719 COMPUTER VISION — FINAL REPORT

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ABSTRACT

The Sequential Keypoint Density Estimator (SeeKer/SKDE, ICCV 2025) is a deliberately simple yet strong baseline for skeleton-based video anomaly detection: it models a pedestrian's 2D keypoint trajectory with an autoregressive density and flags low-likelihood motion. It is, however, *environmentally blind* — each skeleton is scored in isolation, so a pedestrian walking calmly and the same pedestrian walking into a vehicle's path receive identical scores. We extend SeeKer to the *multi-agent* scene. Each vehicle is encoded as a compact oriented-keypoint set (an oriented bounding box rendered as four corners, a centre and a heading), and is either injected as a per-frame context (with an optional FiLM covariance gate) or, in our headline *Scene-SKDE*, concatenated with the pedestrian into a single skeleton of N'=18+6M keypoints modelled under one autoregressive factorisation. On the ShanghaiTech Campus dataset, the context model improves the baseline by +6.0 percentage points of mean frame-AUROC over five seeds (0.726 → 0.786, highly significant). More importantly, a *counterfactual* probe — re-scoring with the vehicle "removed" — shows the gain is a configuration-dependent pedestrian–vehicle *interaction* signal that the pedestrian-only baseline structurally cannot produce (near-vehicle-hazard AUROC 0.79 → 0.97; score–proximity correlation 0.16 → 0.70), rather than mere object presence. We also report honestly: the overall-AUROC advantage narrows under temporal score smoothing, and a full presence-vs-interaction separation requires a benchmark where vehicles are sometimes normal. Our contribution is the multi-agent generalisation of an autoregressive keypoint density and the demonstration of an isolated interaction signal, supported by a reproducible, error-barred evaluation.

1 INTRODUCTION AND PROBLEM STATEMENT

Video anomaly detection (VAD) identifies events that deviate from normal training behaviour. Skeleton-based VAD operates on 2D human poses rather than RGB pixels, gaining low dimensionality, appearance/lighting invariance, and privacy (Morais et al., 2019; Markovitz et al., 2020; Hirschorn & Avidan, 2023). Within this family, SeeKer (Delić et al., 2025) factorises a pedestrian's pose sequence with a masked autoregressive network (MADE, Germain et al., 2015) and scores anomalies by negative log-likelihood (NLL); despite its simplicity it rivals heavier graph- and flow-based models.

The problem. SeeKer conditions only on a pedestrian's *own* past keypoints and aggregates multiple people by a per-frame maximum — it never observes other agents. Yet in traffic-hazard surveillance the anomaly is rarely the gait in isolation; it is the *interaction* — a pedestrian too close to, or in the path of, a vehicle. A motion-only density cannot represent this: the two situations above are equally likely under it. We ask whether SeeKer's elegant keypoint-density formulation can be extended to model the *scene* — pedestrians *and* vehicles — so that one likelihood-based score reflects pedestrian–vehicle interaction.

Contributions. We frame our claims around what the evidence supports.

- **Scene-SKDE: a multi-agent keypoint density.** We generalise SeeKer from a single human skeleton to a joint scene of road agents. Each vehicle becomes a compact oriented-keypoint set and is modelled *together* with the pedestrian under one autoregressive factorisation, with the pedestrian predicted *given* the surrounding vehicles. To our knowledge this is the first multi-agent extension of this estimator (Section 3).
- **Context-conditioned variant with FiLM covariance.** A lighter alternative injects the vehicle geometry as a per-frame context through an autoregressive-preserving shortcut and a FiLM head (Perez et al., 2018) that lets the context modulate the predicted covariance.
- **Evidence of interaction, not object presence.** A counterfactual probe isolates the vehicle's effect on the pedestrian likelihood and shows it is a configuration-dependent interaction signal (Section 6).
- **Honest, reproducible evaluation.** Multi-seed mean±std, a hazard-subset analysis, a careful operating-point study, and a frank limitations section.

2 Related Work

Skeleton-based VAD. Morais et al. (2019) introduced recurrent message passing over joints; Markovitz et al. (2020) clustered spatio-temporal graph embeddings; STG-NF (Hirschorn & Avidan, 2023) fits a normalizing flow over joint graphs. SeeKer (Delić et al., 2025) shows a masked autoregressive density (MADE, Germain et al., 2015) is a strong, overlooked baseline; we build directly on it. Exact-likelihood flows (RealNVP Dinh et al., 2017, Glow Kingma & Dhariwal, 2018) are the natural route to a more expressive conditional (Section 10).

Context- and interaction-aware VAD. A separate strand injects scene/object cues: Ionescu et al. (2019) score per-object features with one-class SVMs. Closest to us are interaction methods: Wiederer et al. (2022) model a density over multiple road-agent trajectories and flag low-density configurations, and ComplexVAD (Doshi & Yilmaz, 2025) targets "complex" anomalies arising from the interaction of two or more actors. These operate on trajectory/graph features. Our mechanism is orthogonal: we keep SeeKer's autoregressive *keypoint* density and widen its support to multiple, heterogeneous agents within one factorisation. The vehicle representation is an oriented bounding box rendered as keypoints — not an articulated skeleton; we use "skeleton" only by analogy.

3 Method

3.1 Baseline: SeeKer in brief

A track is a sequence of frames, each with V=18 COCO-18 2D keypoints. Let $X_{t,n} \in \mathbb{R}^2$ be keypoint n of frame t, $X_{t,<n}$ the earlier keypoints of the same frame, and X_Δ all keypoints of the previous Δ frames. SeeKer factorises

$$p_\theta(X_t \mid X_\Delta) = \prod_{n=1}^{V} p_\theta(X_{t,n} \mid X_{t,<n}, X_\Delta)\,, \quad p_\theta(X_{t,n} \mid \cdot) = \mathcal{N}(\mu_\theta, \Sigma_\theta). \tag{1}$$

Following SeeKer's headline configuration (and the released code), Σ_θ is *diagonal*; a single masked network outputs μ_θ and a per-axis log-variance. (SeeKer ablates a full 2×2 Cholesky covariance but reports it on par with diagonal, 77.8 vs 77.9 on UBnormal, so we keep the diagonal form.) The anomaly score is the confidence-weighted NLL $s(X_t) = -\sum_n c_{t,n} \ln p_\theta(X_{t,n} \mid X_{t,<n}, X_\Delta)$, and the per-frame score is the maximum over people — i.e. people are modelled *independently*, with no object or interaction terms.

3.2 VEHICLES AS ORIENTED KEYPOINTS

For each frame we run YOLOv11 segmentation on the raw image and keep the hazard classes (`car, bus, truck, motorcycle, bicycle`). From each instance mask we fit an oriented rectangle (`cv2.minAreaRect`) and read off K_v=6 keypoints: the four oriented corners, the centre, and a heading point along the major axis. Crucially these are computed in *pixel* space and then normalised, so orientation is faithful (rotating an oriented box in anisotropically-normalised coordinates would distort angles).

Coordinate-frame consistency. A subtle but decisive detail: the pedestrian reference used to make the representation translation/scale-invariant must live in the *same* frame as the detections. SeeKer normalises each pose segment to zero mean and unit variance, which destroys absolute image position; computing the pedestrian centroid from that normalised pose makes the context *pedestrian-independent*. We instead take the centroid from the raw pixel keypoints divided by the frame size, placing it in the same $[0, 1]$ image frame as the YOLO boxes. Each vehicle keypoint is then expressed relative to the pedestrian centroid and divided by the pedestrian's scale.

3.3 TWO WAYS TO USE THE VEHICLES

(a) Context-conditioned CCSKDE. We append the per-frame vehicle representation C_t (either a per-class proximity vector, or the oriented "car matrix" of Section 3.2) to MADE's input as an *unconditional shortcut*: its columns in the input mask are all-ones, so every hidden unit may read every entry of C_t while the keypoint-to-keypoint causality is bit-for-bit preserved. A FiLM head (Perez et al., 2018) maps C_t to per-coordinate (γ, β) that gate the predicted log-variance, $\log \sigma^2 \leftarrow (1+\gamma(C_t)) \log \sigma^2 + \beta(C_t)$, realising a *contextual covariance penalty*; it is zero-initialised, so training begins exactly at the shortcut-only model.

(b) Unified Scene-SKDE. Our headline model treats vehicles as first-class agents. We form one scene skeleton

$$Z_t = [\, V_t^1, \ldots, V_t^M,\ P_t \,] \in \mathbb{R}^{N' \times 2}, \qquad N' = 18 + 6M, \tag{2}$$

concatenating the M nearest vehicle keypoint sets V_t^m with the pedestrian P_t, vehicles ordered *first*, and normalised *jointly*. SeeKer's factorisation (Eq. 1) is applied verbatim to Z, giving one scene density and one hazard score

$$S(t) = -\sum_{n=1}^{N'} c_{t,n} \, \ln p_\theta(Z_{t,n} \mid Z_{t,<n}, Z_\Delta) \, . \tag{3}$$

Because vehicles precede the pedestrian in the order, each pedestrian keypoint is predicted *given* the surrounding vehicles — the interaction is captured by the conditioning. (An ablation in Section 7 shows that, for the unified model, it is the joint inclusion of the vehicles more than this specific ordering that drives the effect.) For frame-level comparability we read out the pedestrian keypoint terms (the last 18), which already reflect the vehicle conditioning. The masked network is the same MADE with its keypoint count generalised from 18 to N'. We verify by an autograd Jacobian sweep that no keypoint depends on an equal- or later-ordered keypoint of the same frame (**0** forbidden-region violations), while pedestrian keypoints *can* attend to vehicle keypoints — the interaction pathway.

Scope of the density. Conditioning is first-order (pedestrian-given-vehicles) and, with a diagonal Gaussian, each conditional is unimodal; genuinely multimodal normal behaviour (passing either side of a parked car) is only approximated. We revisit this in Section 8.

4 DATASETS AND EXPLORATORY ANALYSIS

We use ShanghaiTech Campus (Liu et al., 2018): 13 outdoor scenes, 330 training and 107 test videos. Anomalies (cycling/driving in pedestrian zones, etc.) are typically pedestrian–object interactions, making it the natural benchmark for our hypothesis. Poses are AlphaPose (Fang et al., 2017) COCO-17 keypoints converted to COCO-18 (we use the STG-NF release for reproducibility);

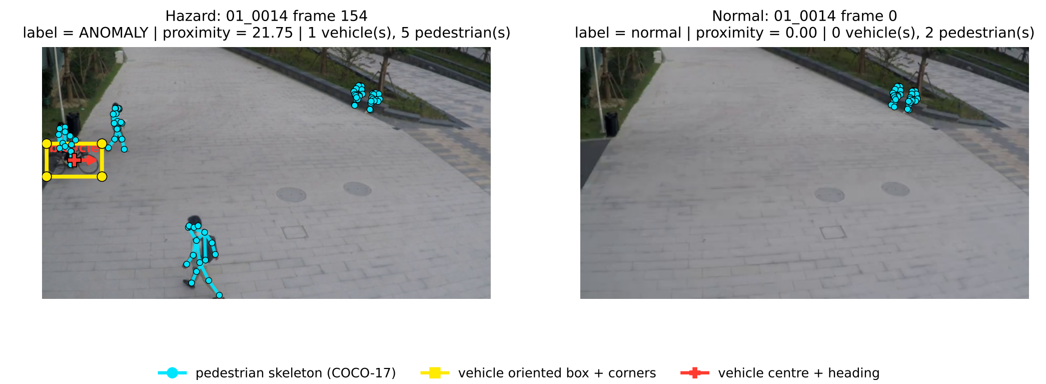

Figure 1: **The scene representation on real ShanghaiTech test frames.** *Left:* an anomalous near-vehicle frame — the pedestrian skeleton (cyan, COCO-17) sits directly against the vehicle's oriented-keypoint set (yellow box corners; red centre and heading arrow), the pedestrian–vehicle *configuration* our Scene-SKDE conditions on. *Right:* a normal frame from the same scene with no vehicle present. The motion-only baseline sees only the cyan skeletons and scores both situations identically; Scene-SKDE additionally conditions the pedestrian keypoints on the yellow/red vehicle keypoints.

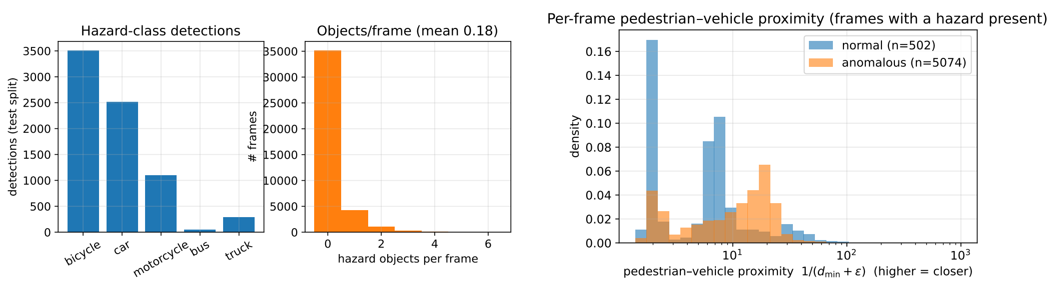

Figure 2: **Dataset EDA.** *Left:* hazard-class detection counts and the distribution of detected objects per frame (most frames contain none — vehicles are rare on a pedestrian campus). *Right:* the distribution of per-frame pedestrian–vehicle proximity $1/(d_{\min}+\epsilon)$ for frames where a hazard is present, split by label; anomalous frames concentrate at *higher* proximity (closer pedestrian–vehicle configurations).

vehicles are obtained once, offline, with YOLOv11 (Ultralytics, 2024) segmentation. The test split comprises $40{,}791$ frames, of which $17{,}326$ (42.5%) are anomalous and $5{,}576$ contain at least one detected hazard.

Figures 2–3 summarise three observations that shape our evaluation: (i) hazard objects are *rare* (most frames have none), so a detector that simply reacts to vehicle presence already correlates with the label; (ii) when a hazard is present, anomalous frames sit at closer pedestrian–vehicle distances; and (iii) near-vehicle frames are $\sim 91\%$ anomalous, a near-degenerate subset that makes a naive "within-interaction" AUROC unstable and motivates the class-balanced and counterfactual metrics below.

5 Experimental Setup

All models train on normal sequences only, minimising the NLL of Eq. 1/3: 10 epochs, batch 1024, segment length 24, stride 1, Adam at $5{\times}10^{-4}$, a single-layer MADE (expansion 1), on a single NVIDIA RTX 5080. Scoring is unchanged from SeeKer (confidence-weighted per-keypoint NLL, max over people, temporal smoothing σ). We report frame-level AUROC, AUPRC and EER, a *per-scene* macro-AUROC, and — because overall AUROC mixes motion-intrinsic with interaction

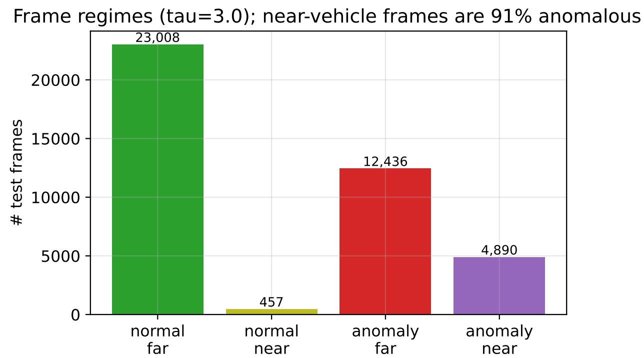

Figure 3: **Frame-regime breakdown.** Among the few frames where a vehicle is near a pedestrian, the overwhelming majority are *anomalous* ($\sim$91%). On ShanghaiTech a vehicle near a pedestrian almost *is* the anomaly — which both motivates a hazard-focused evaluation and warns that this benchmark cannot, by itself, separate "hazard" from "object presence" (Section 8).

Table 1: ShanghaiTech validation AUROC, single seed (10 epochs). All rows share one reproduced baseline and one evaluation pipeline.

Model	Best AUROC	Mean AUROC
SeeKer baseline	0.7351	0.7204
CCSKDE proximity (coord-corrected)	**0.8034**	0.7857
CCSKDE car-matrix (oriented vehicle kpts)	0.8023	0.7855
CCSKDE car-matrix + FiLM	0.7971	**0.7898**
car-matrix, shuffled context (control)	0.7893	0.7835
Unified Scene-SKDE	0.7980	0.7863

anomalies — a hazard-focused suite: a class-balanced near-vehicle AUROC (bootstrap CI), the Spearman correlation between score and pedestrian–vehicle proximity, and a counterfactual interaction probe (Section 6). For significance we use both a seed-level t-test (five seeds $\{42, 1, 2, 3, 4\}$) and, on identical test frames, DeLong's correlated-AUC test with a *clip-level* block bootstrap that respects within-clip temporal correlation.

6 Results

Overview (single seed). Table 1 and Figure 4 compare all variants. Every context/scene variant lifts the baseline by $+6$–7 pp of *mean* AUROC. The corrected proximity context and the richer oriented car-matrix perform comparably (0.803 vs 0.802 best); FiLM gives the best mean (0.790); and the unified Scene-SKDE matches them (0.798 best) through a single, more principled formulation.

A fuller metric suite. Frame-AUROC alone is a thin summary on a 42.5%-positive test set, so we additionally report Average Precision (AUPRC), the Equal Error Rate (EER), and a *per-scene* macro-AUROC (the unweighted mean over ShanghaiTech's 13 scenes; SeeKer reports only the pooled micro value), all scored through the identical pipeline at $\sigma=0$ (Table 2). Every context/scene variant improves *all four* frame-level metrics over the baseline: AUPRC rises by $+8$–11 pp, EER falls by up to 6.5 pp, and the per-scene macro-AUROC rises by $+3$–7 pp — the oriented car-matrix gives the largest macro gain ($0.699 \to 0.770$), evidence that the structured geometry helps *across* scenes rather than over-fitting one. The hazard-specific columns (vehicle-hazard AUROC and score–proximity ρ) widen the gap much further, previewing the interaction analysis below.

Multi-seed significance. Single-seed "best" values are noisy: Table 3 and Figure 5 show the baseline's per-seed best spans 0.735–0.785 (± 0.018), whereas its per-epoch *mean* is a tight 0.7251 ± 0.0042. We therefore compare on the mean. The car-matrix model reaches 0.7856 ± 0.0017:

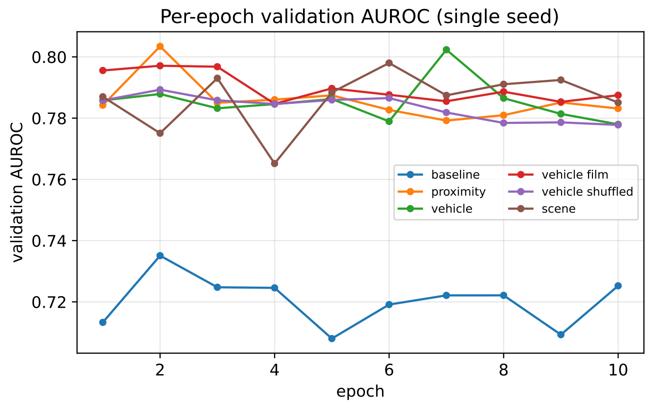

Figure 4: **Per-epoch validation AUROC** (single seed). The baseline (lowest band) is separated from all context/scene variants; the shuffled-context control sits just below the aligned car-matrix model.

Table 2: Frame-level metric suite on ShanghaiTech test (single seed, σ=0, joint read-out for scene). $\uparrow$ higher better, $\downarrow$ lower better. "macro" is the mean per-scene AUROC over 13 scenes; "veh-haz" is the vehicle-hazard subset AUROC; ρ is the Spearman score–proximity correlation.

Model	AUROC↑	AUPRC↑	EER↓	macro↑	veh-haz↑	$\rho\uparrow$
SeeKer baseline	0.735	0.611	0.335	0.699	0.788	0.164
CCSKDE proximity	0.803	0.719	0.279	0.751	0.952	0.583
CCSKDE car-matrix	0.802	0.720	**0.270**	**0.770**	0.950	0.593
CCSKDE car-matrix + FiLM	0.797	**0.726**	0.286	0.768	**0.966**	**0.640**
shuffled control	0.789	0.719	0.294	0.763	0.952	0.602
Unified Scene-SKDE	0.782	0.690	0.301	0.729	0.952	0.601

a $+6.05$ pp improvement, more than an order of magnitude larger than the pooled standard deviation ($\sim$0.4 pp), i.e. highly significant (two-sample t, $p < 10^{-4}$). *This is the headline quantitative result: with error bars, context significantly improves mean AUROC over baseline.*

Paired significance on identical frames. The seed-level t-test above treats whole training runs as the unit. A complementary, stronger statement compares two models on the *same* test frames. We apply DeLong's test for correlated ROC areas and, because frames within a clip are temporally correlated (so a per-frame test overstates significance), a *clip-level* block bootstrap that resamples whole clips — the honest independent unit (Table 4). Every context/scene model beats the baseline with the clip-level 95% CI of the AUROC gain *excluding zero*: $+0.067\,[0.042, 0.095]$ for the car-matrix and $+0.047\,[0.014, 0.082]$ even for the unified Scene-SKDE. The clip-level intervals are $\sim$3–4$\times$ wider than the frame-level ones, as expected, and remain firmly positive.

Interaction vs. mere presence (primary result). Overall AUROC understates the contribution because it mixes anomaly types. We isolate the interaction with a counterfactual: score each frame with the real context and again with the context zeroed (vehicle "removed"), then take the difference (Figure 5, right). Zeroing collapses the model to near-baseline; the interaction term raises near-vehicle-hazard AUROC from 0.79 to $\mathbf{0.97}$ and the score–proximity Spearman correlation from 0.16 to $\mathbf{0.70}$, while being only a weak general detector. A class-balanced within-near-vehicle AUROC (bootstrap) of $\mathbf{0.62}\,[\mathbf{0.59}, \mathbf{0.66}]$ confirms above-chance separation even after removing the $\sim$91% label imbalance (Table 5). Together these say the model fires on the *configuration*, not simply on the presence of a car.

Qualitative demo: predictions in action. Figure 6 traces both models' per-frame score on a single test clip in which a cyclist (a hazard class) rides through a pedestrian entrance. Outside the ground-truth anomaly window both models sit near zero (frames 48, 233). As the hazard be-

Table 3: Multi-seed AUROC, mean±std over 5 seeds. "Best" is the per-seed best-epoch (high variance); "Mean" is the per-seed epoch-mean (stable).

Model	Best AUROC	Mean AUROC
SeeKer baseline	0.756 ± 0.018	0.725 ± 0.004
CCSKDE car-matrix	0.799 ± 0.003	$\mathbf{0.786 \pm 0.002}$

Table 4: Paired significance of the AUROC gain over the baseline on identical test frames (σ=0). DeLong p assumes independent frames (optimistic); the clip-level block bootstrap (107 clips, 2000 resamples) is the honest unit. Both agree the gains are significant.

Comparison (vs. baseline)	ΔAUROC	clip-level 95% CI	clip-boot p
+ proximity	+0.068	[+0.045, +0.096]	< 0.0005
+ car-matrix	+0.067	[+0.042, +0.095]	< 0.0005
+ car-matrix + FiLM	+0.062	[+0.036, +0.092]	< 0.0005
+ Scene-SKDE	+0.047	[+0.014, +0.082]	0.0015

gins the car-matrix score jumps to its maximum and *stays* elevated through the vehicle-present interval (green ticks), whereas the baseline rises at the onset but decays mid-event; at the vehicle-present peak (frame 180) the oriented bicycle keypoints sit beside the pedestrians and our model scores 0.87 versus the baseline's 0.49. This is the intended behaviour — the score responds to the pedestrian–vehicle *configuration*, not to motion alone. An animated version is provided with the code (`demo_detection.gif`).

Operating-point sensitivity (an honest caveat). ShanghaiTech AUROC is sensitive to temporal smoothing σ (Figure 7). Smoothing lifts the baseline ($0.781 \rightarrow 0.797$ at σ=20) but *lowers* the car-matrix model ($0.802 \rightarrow 0.765$), because the interaction signal is temporally punctate. At the baseline's best operating point the *overall*-AUROC gap narrows to ∼1 pp. We report this rather than tuning σ in our favour: the defensible claim is *comparable overall AUROC plus a distinct interaction signal*, not a global AUROC win.

Reproduction gap. Our baseline (mean 0.725, best 0.797 with smoothing) sits below SeeKer's published 0.855. We attribute the residual to (i) the public code path used here scores only the last frame of each window via `validate`; the full-window test-time scorer (`joint_lnp_full_window`) that `test` calls is *undefined* in the released snapshot, and (ii) unspecified training/scoring details common to skeleton-VAD reproductions. All our comparisons use one reproduced baseline and pipeline, so relative claims are unaffected.

Efficiency. The density models are lightweight: 2.24/2.97/6.22 M parameters (baseline/car-matrix+FiLM/scene), 0.30/0.41/0.72 ms per batch ($> 3 \times 10^7$ frame-scores/s on one RTX 5080). YOLO/AlphaPose are one-time offline costs.

7 Ablation Studies

Beyond the variant comparison of Tables 1–2, we run two controlled sweeps that isolate specific design choices (Table 6, σ=0, single seed): the number of vehicles M kept per frame, and the agent ordering in the unified model.

(i) Context value: adding any context lifts mean AUROC by +6–7 pp and improves all four frame-level metrics (Table 2). **(ii) Number of vehicles:** M=2 is the sweet spot; M=1 is close and M=3 is slightly *worse* on overall AUROC, AUPRC and per-scene macro — on a pedestrian campus a third nearby vehicle is rare, so the extra slot mostly contributes zero-fill noise. **(iii) Agent ordering:** placing vehicles before vs. after the pedestrian changes *every* metric by ≤ 0.3 pp — a near-wash. The honest reading is that for the unified model it is the *joint inclusion* of vehicles in the density, more than the specific autoregressive order, that carries the effect; the conditioning direction is a second-order factor once the aggregate scene likelihood already sees the vehicle keypoints. **(iv) Spatial**

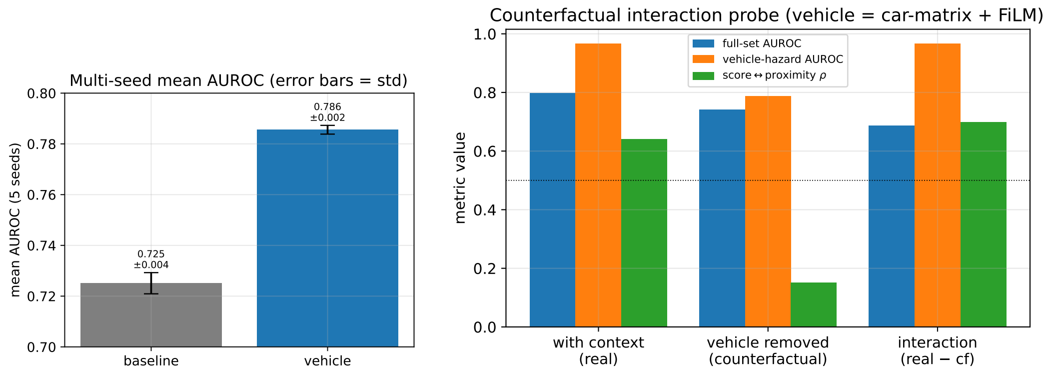

Figure 5: **Left:** multi-seed mean AUROC (error bars = std over 5 seeds); the baseline/car-matrix gap is large relative to the spread. **Right:** counterfactual interaction probe. Removing the vehicle (zeroing the context) collapses the model to near-baseline (full 0.74, vehicle-hazard 0.79, ρ=0.15); the isolated interaction term (real−cf) is a near-perfect *vehicle-hazard* detector (0.97, ρ=0.70) yet a weak *general* detector (0.69) — the signature of interaction, not constant presence.

Table 5: Hazard-focused metrics (car-matrix + FiLM, σ=0). The "interaction" row is the counterfactual difference (real − vehicle-removed).

Variant	full-set AUROC	vehicle-hazard AUROC	score–proximity ρ
SeeKer baseline	0.735	0.79	0.16
with context (real)	0.797	0.966	0.640
vehicle removed (counterfactual)	0.741	0.788	0.151
interaction (real − cf)	0.687	**0.967**	**0.698**
balanced near-vehicle AUROC (bootstrap): 0.62 $[0.59, 0.66]$			

alignment (shuffled control): permuting the context across segments — identical parameter count, geometry misaligned — drops the car-matrix best from 0.802 to 0.789 and weakens every hazard metric (Table 2); the gap is the portion attributable to genuine spatial alignment, the remainder to the regularising effect of a richer input. **(v) FiLM covariance:** the FiLM gate yields the best AUPRC, the strongest vehicle-hazard AUROC and the strongest score–proximity coupling (Table 2). **(vi) Representation:** the oriented car-matrix and the simple proximity vector are comparable on overall AUROC, but the oriented geometry gives a clearly better per-scene macro (0.770 vs 0.751), i.e. geometry aids cross-scene generalisation. **(vii) Temporal smoothing** (Figure 7) remains the most influential test-time hyper-parameter.

8 Limitations

(1) Single dataset. We evaluate only on ShanghaiTech; SeeKer also uses UBnormal and MSAD-HR. **(2) Presence vs. interaction not fully disentangled.** On ShanghaiTech a near-vehicle frame is $\sim$91% anomalous (Fig. 3); the counterfactual is strong evidence of configuration-dependence, but a definitive separation needs a benchmark where vehicles are sometimes normal (e.g. MSAD-HR). **(3) Overall-AUROC gain is operating-point-sensitive** (Fig. 7); we therefore foreground the interaction metrics. **(4) Reproduction gap** to the published 0.855. **(5) Model capacity:** a first-order, diagonal-Gaussian, single-layer MADE cannot represent higher-order multi-agent dynamics or multimodal normal motion. **(6) Scene read-out:** the unified model is scored on pedestrian keypoints for comparability, leaving vehicle-intrinsic hazards to a future joint read-out.

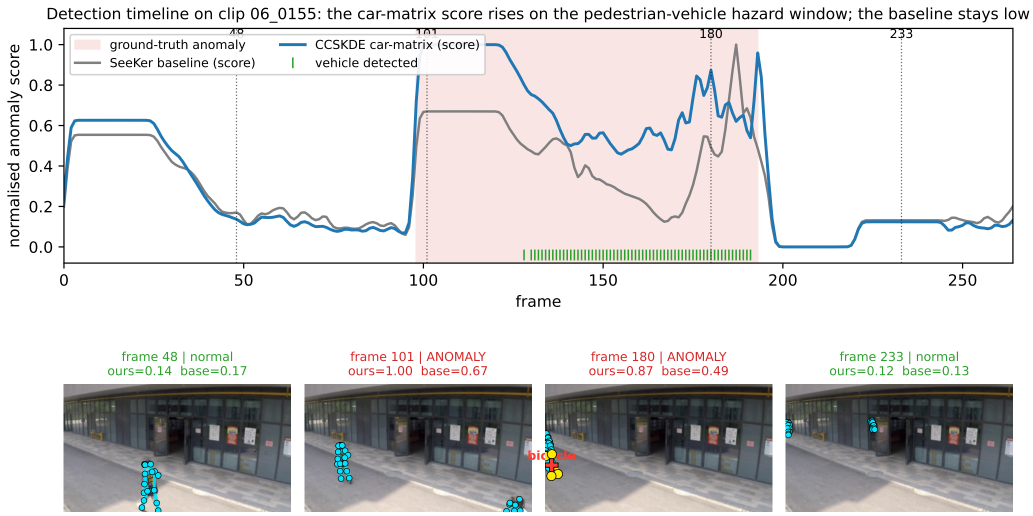

Figure 6: **Detection demo on clip 06_0155.** *Top:* normalised per-frame anomaly score for the SeeKer baseline (grey) and our CCSKDE car-matrix model (blue); the ground-truth anomaly window is shaded, green ticks mark frames with a detected hazard vehicle. *Bottom:* keyframes with the pedestrian skeleton (cyan) and oriented vehicle keypoints (yellow/red), annotated with each model's score. Our model stays high across the hazard; the baseline lags and decays.

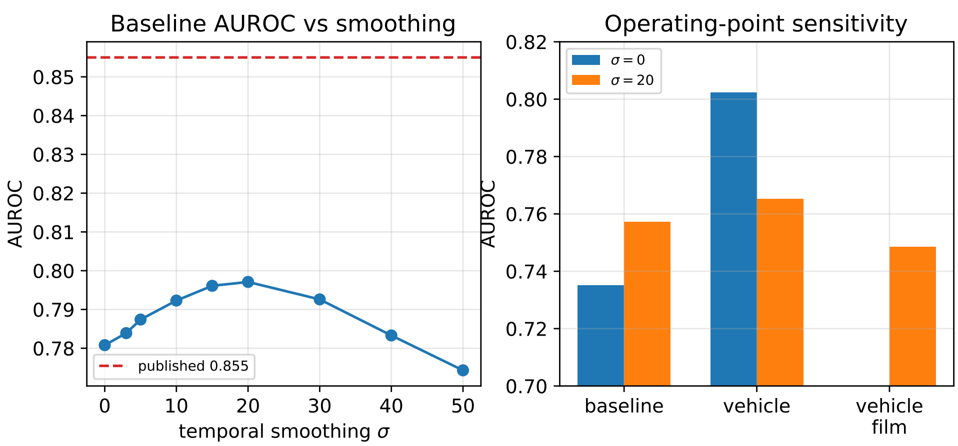

Figure 7: **Reproduction gap and operating-point sensitivity.** *Left:* baseline AUROC vs smoothing σ; smoothing accounts for $\sim$1.6 pp, leaving a residual gap to the published 0.855 (see text). *Right:* at σ=20 the overall-AUROC advantage of the context model nearly vanishes.

9 Discussion

The clean finding is that an autoregressive keypoint density *can* be extended to multiple, heterogeneous agents without abandoning SeeKer's mechanism or efficiency, and that doing so injects a measurable pedestrian–vehicle interaction signal — exactly the quantity a motion-only model is blind to. The counterfactual analysis is, in our view, more convincing than a raw AUROC delta: it shows the contribution is the *vehicle's effect on the pedestrian likelihood*, which behaves like interaction (configuration-dependent, strongly proximity-coupled) rather than a presence prior. At the same time, the operating-point study is a cautionary tale for skeleton-VAD comparisons: an unsmoothed AUROC can flatter a model whose signal is temporally punctate. We deliberately report the less favourable smoothed comparison.

Table 6: Controlled ablations (σ=0, single seed). *Top:* number of vehicles M in the car-matrix context. *Bottom:* agent ordering in the unified Scene-SKDE (vehicles-first = pedestrian conditioned on vehicles; pedestrian-first removes that conditioning at identical capacity).

Ablation	AUROC	AUPRC	EER↓	per-scene macro	veh-haz
Number of vehicles M (car-matrix context)					
M=1	0.799	0.720	0.283	0.757	0.958
M=2 (default)	**0.802**	0.720	**0.270**	**0.770**	0.950
M=3	0.788	0.711	0.298	0.747	0.957
Agent ordering (unified Scene-SKDE, joint read-out)					
vehicles-first (default)	0.782	0.690	0.301	0.729	0.952
pedestrian-first	0.779	0.684	0.300	0.726	0.949

10 Future Directions

(1) **MSAD-HR / cross-dataset:** a setting with normal vehicles would separate hazard from presence and test generalisation. (2) **Full multi-seed** of every config with significance tests. (3) **Expressive conditional:** replace the diagonal Gaussian with a normalizing-flow or diffusion head (SeeKer's own ablation shows richer *covariance* buys little, so expressivity should come from the conditional). (4) **Universal keypoint agents:** a promptable keypoint model (e.g. X-Pose) would let any human, animal, or object become an agent in the same density — a path to "SKDE for anything with keypoints", including animal-behaviour and crowd anomalies. (5) **Hazard anticipation:** roll out the density to score predicted near-collisions (a learned TTC/PET surrogate).

References

Delić, A., Grčić, M., & Šegvić, S. (2025). Sequential Keypoint Density Estimator: An Overlooked Baseline of Skeleton-Based Video Anomaly Detection. In *ICCV* (Highlight).

Germain, M., Gregor, K., Murray, I., & Larochelle, H. (2015). MADE: Masked Autoencoder for Distribution Estimation. In *ICML*.

Hirschorn, O., & Avidan, S. (2023). Normalizing Flows for Human Pose Anomaly Detection. In *ICCV*.

Morais, R., Le, V., Tran, T., Saha, B., Mansour, M., & Venkatesh, S. (2019). Learning Regularity in Skeleton Trajectories for Anomaly Detection in Videos. In *CVPR*.

Markovitz, A., Sharir, G., Friedman, I., Zelnik-Manor, L., & Avidan, S. (2020). Graph Embedded Pose Clustering for Anomaly Detection. In *CVPR*.

Ionescu, R. T., Khan, F. S., Georgescu, M.-I., & Shao, L. (2019). Object-Centric Auto-Encoders and Dummy Anomalies for Abnormal Event Detection in Video. In *CVPR*.

Doshi, K., & Yilmaz, Y. (2025). ComplexVAD: Detecting Interaction Anomalies in Video. In *WACV*.

Wiederer, J., Bouazizi, A., Troina, M., Kressel, U., & Belagiannis, V. (2022). Anomaly Detection in Multi-Agent Trajectories for Automated Driving. In *CoRL*.

Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. (2018). FiLM: Visual Reasoning with a General Conditioning Layer. In *AAAI*.

Dinh, L., Sohl-Dickstein, J., & Bengio, S. (2017). Density Estimation Using Real NVP. In *ICLR*.

Kingma, D. P., & Dhariwal, P. (2018). Glow: Generative Flow with Invertible $1{\times}1$ Convolutions. In *NeurIPS*.

Liu, W., Luo, W., Lian, D., & Gao, S. (2018). Future Frame Prediction for Anomaly Detection – A New Baseline. In *CVPR*.

Fang, H.-S., Xie, S., Tai, Y.-W., & Lu, C. (2017). RMPE: Regional Multi-Person Pose Estimation. In *ICCV*.

Ultralytics. (2024). YOLOv11. `https://github.com/ultralytics/ultralytics`.